CELL PHONE DROP TEST

Overview

This document simulates the process of a cell phone dropping and impacting the ground. The simulation aims at illustrating the idea of strain and how sometimes it is a more useful measurement than deformation.

Goals

- To understand the difference between strain and deformation

Steps

1. Create an "explicit dynamics" analysis system in Ansys workbench.
2. Define materials. Define an Aluminum alloy material for the cellphone. The purpose of this simulation is to illustrate the idea of strain only. Thus we assume the material to be aluminum and do not pursue accuracy.
3. Import the geometry (Figure 1).

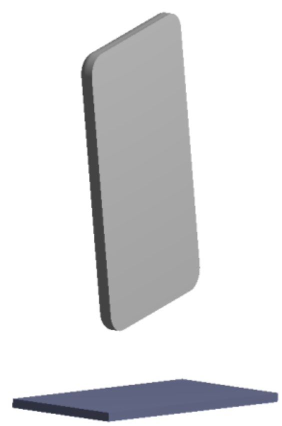
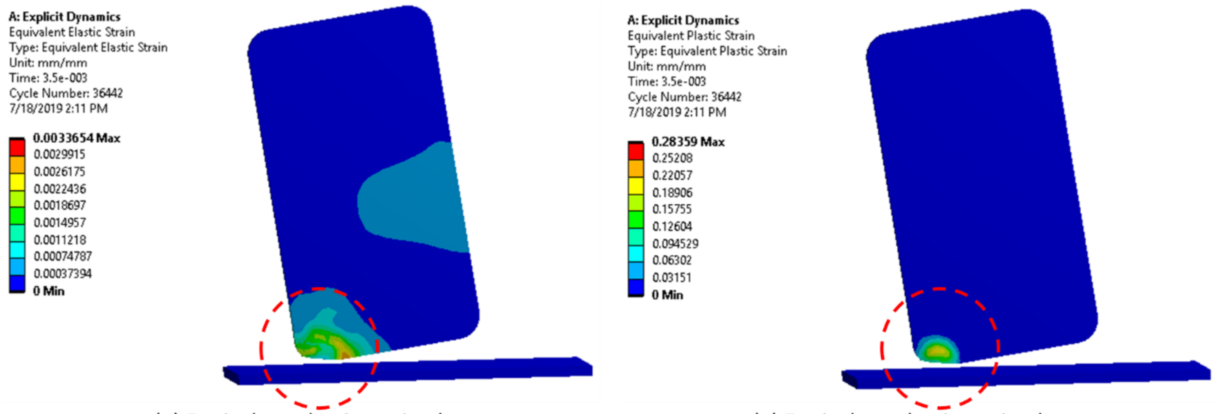


Figure 1. The geometry of cellphone

4. Assign material to geometries. Change the ground part to rigid.
5. Define body interaction and mesh the geometry.
6. Define analysis settings and boundary conditions. Fix the part that represents ground and applies a standard earth gravity. Add an initial velocity of 20m/s (pointing downwards) to the cellphone. Note that this high-velocity value is to achieve a short computational time and not realistic. Define the simulation end time to be 0.035s.
7. Run the simulation and review the results. A contour plot of the equivalent elastic and plastic strain at time 0.035s are shown in Figure 2. The largest strain happens in the red circular region. From the deformation plot at time 0.035s (Figure 3), however, the largest deformation occurs at the right edge. It can be observed that large deformation does not necessarily

produce large strain, as deformation is a combination of rigid body movement and strain.



(a) Equivalent elastic strain plot (a) Equivalent plastic strain plot

Figure 2. Equivalent elastic strain contour plot

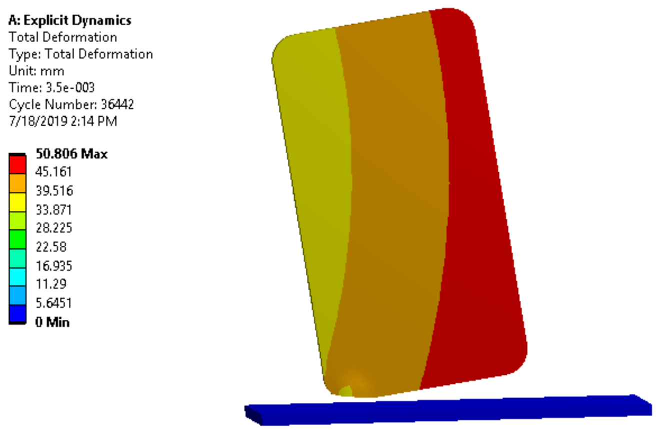


Figure 3. Deformation contour plot

Summary

This document uses an explicit dynamic analysis of cellphone drop simulation to illustrate the difference between deformation and strain.